

Practice Problems 10

Parallel light of wavelength $6563 \text{ \AA}$ is incident normally on a slit 0.3850 mm wide. A lens with a focal length of 50.0 cm is located just behind the slit bringing the diffraction pattern to focus on a white screen. Find the distance from the center of the principle maximum to (a) the first minimum and (b) the fifth minimum.

Ans. (a) 0.852 mm , (b) 4.261 mm

Calculate the approximate intensity of (a) the first and (b) the second weak maxima that appear along the diagonal $\beta/\gamma = l/b$ in the Fraunhofer diffraction pattern of a rectangular aperture of width b and height l .

Ans. (a) I/I_0 0.2227% , (b) I/I_0 0.02716%